DIVINE GRACE COLLEGE OF ARTS, SCIENCE AND TECHNOLOGY (DGCAST) – GAROUA

GEOGRAPHY REVISION SHEET – FORM THREE

First Sequence Evaluation, School Year 2026-2027 • Teacher: Mr KAMDEM Emmanuel

Lessons revised: Our Planet; Universe and Moon; Latitudes and Longitudes; Great Circles and IDL; Locate a Place on a Map

A. OUR PLANET, THE EARTH

1. What are the size and the shape of the Earth?

Answer: The Earth is an **oblate spheroid**: round, slightly flattened at the poles and bulging at the Equator. Equatorial circumference about 40,075 km; polar circumference about 40,008 km; equatorial diameter about 12,756 km; polar diameter about 12,714 km; surface area about 510 million km².

2. Give four proofs that the Earth is spherical.

Answer: A ship sailing away disappears bottom first; the shadow of the Earth on the Moon during an eclipse is always curved; photos from space show a round Earth; a traveller who always goes in one direction comes back to the starting point; the horizon is a circle that widens as we go higher; the Sun rises earlier in the east than in the west.

B. FURTHER STUDY 1: THE UNIVERSE AND THE MOON

3. What does the universe contain and how is it linked to the Earth?

Answer: Billions of galaxies (ours is the **Milky Way**), stars (the Sun is the nearest), the solar system, asteroids, comets, gases and dust. Links: the Sun gives light and heat; the Earth moves round the Sun (year and seasons); the Sun and the Moon pull the Earth (tides); the Earth is the third planet and the only one with life.

4. Name the phases of the Moon. Give impacts of the Moon on the environment and on human activities.

Answer: Month of about 29½ days: new Moon, first quarter, full Moon, last quarter.
Environment: tides (two a day); spring tides (new and full Moon) and neap tides (quarters); light at night; eclipses.
Human activities: fishing with the tides, big ships entering shallow ports such as Douala at high tide, calendars and feasts (Ramadan), planting according to the Moon.

C. LATITUDES AND LONGITUDES

5. Define latitude, longitude, parallels, meridians and geographical coordinates.

Answer: Latitude: angular distance north or south of the Equator. **Longitude**: angular distance east or west of the Greenwich Meridian. **Parallels**: lines of equal latitude. **Meridians**: lines of equal longitude, from pole to pole. **Coordinates**: latitude and longitude together.

6. Give the major lines of latitude and longitude with their values.

Answer: Equator 0°; Tropic of Cancer 23½° N; Tropic of Capricorn 23½° S; Arctic Circle 66½° N; Antarctic Circle 66½° S; poles 90° N and S. Greenwich (Prime) Meridian 0°; the 180° meridian is opposite it. One degree of latitude is about 111 km.

7. Give three differences between latitudes and longitudes.

Answer: Latitudes run west–east, never meet and have unequal lengths (shorter towards the poles); longitudes run north–south, meet at the poles and have equal lengths. Latitudes go up to 90°, longitudes up to 180°.

D. GREAT CIRCLES AND THE INTERNATIONAL DATE LINE

8. What is a great circle? Give examples and its importance.

Answer: A circle whose plane passes through the centre of the Earth and cuts it into two equal halves. Examples: the **Equator** (the only parallel that is a great circle) and every meridian joined to its opposite. The tropics and polar circles are small circles. Importance: shortest route between two places, used by aeroplanes and ships.

9. Where is the International Date Line and why is it important?

Answer: It follows the **180° meridian** in the Pacific Ocean and bends around islands and countries. West of it, the date is one day ahead. It shows where each new day begins. Crossing it **westwards**, a traveller adds one day; crossing it **eastwards**, he removes one day.

E. PRACTICAL WORK 1: LOCATE A PLACE ON A MAP

10. How do you read the coordinates of a place on a map?

Answer: Find the place; follow its parallel to the side of the map and read the latitude with N or S; follow its meridian to the top or bottom and read the longitude with E or W; write the latitude first, for example **9° N, 13° E**.

11. Give the position of Cameroon and the coordinates of four towns.

Answer: Cameroon lies between about 2° N and 13° N and between 8° E and 16° E (Central Africa). Garoua 9° N, 13° E; Maroua 11° N, 14° E; Ngaoundéré 7° N, 14° E; Yaoundé 4° N, 12° E.

Quick quiz	Answer
Natural satellite of the Earth	**The Moon**
Tropic of Cancer	**23½° N**

Longitude 0°	**Greenwich Meridian**
The only parallel that is a great circle	**The Equator**

F. PRACTICE OF COMPETENCE

Situation

Your uncle flies from Douala to New York. On the screen in the aeroplane, he sees that the route is curved, not straight, and asks why the pilot does not fly in a straight line.

12. What is the problem raised in the situation?

Answer: Your uncle does not understand why the aeroplane follows a **curved route** on the map: he does not know the notion of great circle.

13. Propose three ways to explain it to him.

Answer: Show him on a globe, with a string, that the shortest line between two places is an arc of a **great circle**; explain that the flat map hides the curve of the Earth; say that aeroplanes and ships follow great circle routes to save time and fuel.

Exam tip: *In Part Two, Question 1 asks for the problem of the situation, Question 2 is answered with* ***Document 1****, and Question 3 asks you to propose your own solutions.* ***Document 2 has nothing to do with the situation: do not use it.***